

Cs601

Data Communication

Most Repeated Subjective Midterm

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Short Question

Question No 1

Write down three characteristics of Sine Wave.

Answer :-

Amplitude

Period/Frequency

Phase

Question No 2

A Sine wave has a frequency of 6 Hz. What is its period?

Answer:

$F = 6$, $t = ?$

$T = 1/f = 1/6 = 0.17 \text{ sec}$

Question No 3

How does noise affect FSK?

Answer:

The duration of the noise is normally longer than the duration of a bit which means that when noise affects data, it affects a set of bits.

Question No 4

What is simplex mode of transmission?

Answer:

Communication is Unidirectional only one of the two stations can transmit other can only receive

Question No 5

How Many Components of a Data Communication system?

Answer:-

1. Message
2. Sender
3. Receiver
4. Transmission Medium
5. Set of rules (Protocol)

Question No 6

What is purpose of distributed processing?

Answer:

Multiple computers working on a problem can solve a problem faster than a computer working alone.

Question No 7

Give methods of Analog to Analog conversion.

Answer:

Analog Conversion:

Representation of analog information by an analog signal is Analog –to- analog conversion.

Question No 8

How many types of Bipolar encoding scheme exist?

Answer:

Bipolar uses three voltage levels.

There are three types of bipolar encoding:

1. AMI (Alternate Mark Inversion)
2. B8ZS
3. HDB3

Question No 9

Define Sampling Rate.

Answer:-

The accuracy of any digital reproduction of an analog signal depends upon the o. of samples taken

Question No 10

How bandwidth of a Modem is measured?

Answer:

Bandwidth of a modem is not measured in bytes, it is measured in bits. Modem Speed –ASK (full-duplex) effective bandwidth of a telephone line is 2400 Hz, the maximum baud rate is also

Question No 11

What is frequency spectrum?

Answer:

Frequency spectrum is the collection of all the components frequencies it contains.

Question No 12

What is digital signal? Give an example

Answer:

A digital signal is discrete. It can have only a limited number of defined values, often as simple as 1s and 0s the transition of a digital signal from value to value is instantaneous like a light being switched ON and OFF.

Question No 13

Give similarities between 4 QAM & 8 QAM.

Answer:

In both case no. of amplitude shifts is more than the no. of phase shifts because amplitude changes are susceptible to Noise, number of phase shifts used by QAM is always larger than the amplitude shift

Question No 14

What is HDB3.

Answer:

Alteration of AMI adopted in Europe and Japan introduces changes into AMI, every time four consecutive zeros are encountered instead of waiting for eight zeros as in the case of B8ZS.

Question No 15

Define Bit interval and Bit rate.

Answer:

Bit Interval and Bit Rate

Most digital signals are aperiodic and thus Period and Frequency are not the appropriate terms to describe them

Bit Interval (seconds)

Time required to send one single bit

Bit Rate (bps)

Number of bits sent per second

Long Question

Question No 1

Describe different types of NRZ encoding scheme?

Answer:

In NRZ encoding the level of signal is either positive or negative. NRZ is

Classified into 2 types:

NRZ-L NRZ-I

NRZ-L:

Level of the signal depends on the type of bit it represents.

Positive voltage usually means the bit is a 1 and a negative voltage usually means the bit is a 0.

NRZ-I:

The inversion of the level represents a 1 bit. A bit 0 is

Represented by no change.

NRZ-I is superior to NRZ-L.

Question No 2

Briefly describe Return to Zero (RZ) encoding scheme?

Answer:

- Any time, data contains long strings of 1's or 0's, Rx can lose its timing.
- In unipolar, we have seen a good solution is to send a separate timing signal but this solution is both expensive and full of error.
- A better solution is to somehow include synch in encoded signal somewhat similar to what we did in
- NRZ-I but it should work for both strings of 0 & 1.
- One solution is RZ encoding which uses 3 values : Positive, Negative and Zero.

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- Signal changes not b/w bits but during each bit.
- Like NRZ-L , +ve voltage means 1 and a –ve voltage means 0, but unlike NRZ- L, half way through each bit interval, the signal returns to zero.
- A 1 bit is represented by positive to zero and a 0 is represented by negative to zero transition.
- The only problem with RZ encoding is that it requires two signal changes to encode one bit and therefore occupies more BANDWIDTH.
- But of the 3 alternatives we have discussed, it is most effective value.

Question No 3

Difference between Phase Modulation(PM) with Frequency Modulation(FM)?

Answer:

Phase modulation (PM)

- Simpler hardware requirements
- Phase is modulated with the amplitude
- Amplitude & Frequency of the carrier signal remain constant

Frequency Modulation (FM)

- Frequency of carrier signal is changed according to the amplitude of modulating signal
- Amplitude and Phase of the carrier signal remain constant

FM Bandwidth

Bandwidth of FM signal (modulated signal) = $10 \times$ bandwidth of modulating signal
Significant spectrum of FM

Audio = 15 KHz

Minimum 150 KHz bandwidth

Question No 4

Write down the names of layers present in TCP/IP Protocol Suite?

Answer:

Physical (physical standards)

Data Link (N/w Interface)

Network (Interconnectivity)

Transport (Transport Functions)

Application (Session, Pres, app of OSI)

Question No 5

Write a note on Full duplex mode of transmission.

Answer:

Both stations can transmit and receive simultaneously

Two way street with traffic flowing in both directions at the same time

Signals traveling in either direction share the capacity of the link

The sharing can take place in two ways:

Either the link must contain two physically separate transmission paths:

- One for sending and
- One for receiving

Question No 6

Characteristics of a Data Communication System?

Answer:-

The effectiveness of a data communications system depends on four fundamental characteristics:

- 1.- The system must deliver data to the correct destination. Data must be received by the receiver or user and only by that device or user.
2. Accuracy- The system must deliver the data accurately.
- 3.- The system must deliver data in a timely manner. Data delivered
4. Late are useless. In the case of video and audio, timely delivery means delivering .

5.Data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.

6.Jitter- Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets

Question No 7

Differentiate between the following terms.

- (a) Syntax and Semantics of protocol.
- (b) Network management and exchange management.
- (c) Monolithic and structured.

Answer

Syntax and Semantics of protocol

Syntax

Communication between peer entities

Semantics:

Client/server (P21)

Network management and exchange management

Exchange Management

Error Detection and Correction

Network management

Security

Monolithic and structured

Monolithic

Communications is a complex task

To complex for single unit

Structured

Structured design breaks down problem into smaller units layered structure

Question No 8

Differentiate between the following terms.

- a. Frequency spectrum and bandwidth.
- b. Bandwidth and throughput.
- c. Bit interval and bit rate.

Answer

Frequency Spectrum / Bandwidth

Frequency Spectrum: of a signal is the collection of all the component frequencies it contains It is shown using a Frequency domain graph

Bandwidth: of a signal is the width of the frequency spectrum in other words, Bandwidth refers to the range of the component frequencies and Frequency Spectrum refers to the elements within that range

Bandwidth and throughput.

Measurement of how fast data can pass through a point o In other words, if we consider any point in the TX Medium as a wall through which bits pass, then throughput is the number of bits that can pass this wall in second

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